

# A Literature-Implied Negative Answer to the BCAP-Free Calkin Embedding Question

FA Banach run `fa_banach_001`, agent `agent_lane_05`

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## Status

This is a lightweight literature-implied answer packet. It records that Remark 1 after Theorem 2.1 of Boedihardjo–Johnson is answered negatively once the lane-5 norm-identification packet is combined with the Astala–Tylli measure-of-noncompactness results.

The conclusion is not claimed as a new counterexample produced by the run. The run contribution is the identification of the Boedihardjo–Johnson representation norm with the measure-of-noncompactness norm, which makes the existing Astala–Tylli separation apply directly.

## Source Question

Boedihardjo and Johnson define, for an infinite-dimensional Banach space  $X$ , an ultrapower representation

$$f : \mathcal{B}(X)/\mathcal{K}(X) \longrightarrow \mathcal{B}(\widehat{X}), \quad \dot{T} \mapsto \widehat{T}.$$

Their Theorem 2.1 proves that if  $X$  has the  $\lambda$ -bounded compact approximation property, then  $f$  is a norm-one  $(\lambda + 1)$ -embedding. Immediately after the theorem they write:

“We do not know whether Theorem 2.1 is true without the hypothesis that  $X$  has the BCAP.”

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It follows that  $\|f(\dot{T})\| \geq (1 + \lambda)^{-1}$  whenever  $\|T\|_e > 1$ . □

**Remark 1.** We do not know whether Theorem 2.1 is true without the hypothesis that  $X$  has the BCAP.

**Remark 2.** The embedding in Theorem 2.1 is an isometry if the approximating net can be chosen so that  $\|I - S_\alpha\| = 1$  for every  $\alpha$ . This is the case if, for example, the space  $X$  has a 1-unconditional basis. However, we do not know whether the embedding is an isometry if  $X = L_p(0, 1)$  with  $p \neq 2$ .

If  $N$  is a subset of  $Y^*$ , then we can define a subset  $N'$  of  $\widehat{Y}$  by

## Identification

For  $T \in \mathcal{B}(X)$ , let

$$\gamma_X(T) = \inf\{\varepsilon > 0 : T(B_X) \subset K + \varepsilon B_X \text{ for some compact } K \subset X\}$$

be the Hausdorff measure of noncompactness of  $T(B_X)$ . Equivalently,

$$\gamma_X(T) = \inf\{\|Q_E T\| : E \subset X, \dim E < \infty\},$$

where  $Q_E : X \rightarrow X/E$  is the quotient map. The equivalence follows because a bounded subset of a finite-dimensional subspace is compact.

The previous lane-5 packet proved the exact norm formula

$$\|\widehat{T}\| = \gamma_X(T) \quad (T \in \mathcal{B}(X)).$$

Thus the BCAP-free version of Theorem 2.1 is equivalent to asking whether, for each Banach space  $X$ , there is  $C_X < \infty$  such that

$$\|T\|_e \leq C_X \gamma_X(T) \quad (T \in \mathcal{B}(X)).$$

The opposite inequality  $\gamma_X(T) \leq \|T\|_e$  is automatic.

## Literature Input

Astala and Tylli studied the bounded compact approximation property through measures of non-compactness [2]. Later sources quote the relevant consequences in exactly the form needed here:

- A later paper recalls that the Hausdorff measure of noncompactness induces a norm on  $\mathcal{B}(X, Y)/\mathcal{K}(X, Y)$ , but that this norm need not be complete in general, citing Astala–Tylli, Theorem 2.5 [4, Subsection “Measures of non-compactness”].
- The same source quotes Lebow–Schechter, supplemented by Astala–Tylli Theorems 2.2 and 2.3, for the estimate

$$\|T\|_e \leq \delta \gamma(T)$$

whenever the range space has the  $\delta$ -compact approximation property [4, Subsection “Measures of non-compactness”].

- Skillicorn records that Zemanek noticed a result of Astala–Tylli providing the first example of a Calkin algebra with a non-unique norm [3, Section 1]; in the surrounding discussion these non-equivalent norms are noted to fail completeness [3, Section 1].

Crossref metadata also records a later direct paper by D. Chen, *Characterizations of bounded compact approximation property by Calkin representations*, Proc. Amer. Math. Soc. 150 (2022), 5397–5402, DOI 10.1090/proc/16056. Its abstract says that it characterizes BCAP via Calkin representations, and its reference list includes both Boedihardjo–Johnson and Astala–Tylli. The AMS PDF was blocked by Cloudflare from this environment, so this packet does not quote Chen’s theorem statement.

## Consequence for the Boedihardjo–Johnson Question

The Astala–Tylli/Skillicorn example gives a Banach space  $X$  for which the measure-of-noncompactness norm  $\gamma_X$  on the Calkin algebra  $\mathcal{B}(X)/\mathcal{K}(X)$  is not equivalent to the usual quotient norm  $\|\cdot\|_e$ . Since always  $\gamma_X(T) \leq \|T\|_e$ , non-equivalence means that no constant  $C_X$  can satisfy

$$\|T\|_e \leq C_X \gamma_X(T) \quad (T \in \mathcal{B}(X)).$$

Equivalently, there is a sequence  $T_n \in \mathcal{B}(X)$  such that

$$\|T_n\|_e = 1, \quad \gamma_X(T_n) \longrightarrow 0.$$

Using the lane-5 formula  $\|\widehat{T}\| = \gamma_X(T)$ , this becomes

$$\|T_n\|_e = 1, \quad \|\widehat{T}_n\| \longrightarrow 0.$$

Therefore the Boedihardjo–Johnson representation is not bounded below on the Calkin algebra of this  $X$ . Hence Theorem 2.1 does not remain true for all Banach spaces without the BCAP hypothesis.

## Scope

This packet answers the all-Banach-spaces form of Remark 1 negatively. It does not identify whether every non-BCAP space fails the representation embedding, nor does it settle the isometry question in Remark 2 for  $L_p(0, 1)$ ,  $p \neq 2$ .

## Verification Notes

Human review should check one external identification: in the Astala–Tylli/Skillicorn example, the non-equivalent Calkin algebra norm is indeed the Hausdorff measure-of-noncompactness quotient norm  $\gamma$ . Once that is confirmed, the argument above is immediate from the previous lane-5 norm-identification packet.

## References

- [1] M. T. Boedihardjo and W. B. Johnson, *On Mean Ergodic Convergence in the Calkin Algebras*, Proc. Amer. Math. Soc. 143 (2015), 2451–2457; arXiv:1301.0054.
- [2] K. Astala and H.-O. Tylli, *On the bounded compact approximation property and measures of noncompactness*, J. Funct. Anal. 70 (1987), 388–401.
- [3] R. Skillicorn, *The uniqueness-of-norm problem for Calkin algebras*, arXiv:1507.08118.
- [4] P. Kolwicz, K. Leznicki, and L. Maligranda, *Essential norm of multiplication operators between Kothe spaces*, arXiv:2212.06723.